

Important Directions

- **Read through all slides**
 - **Watch all videos**

The Newborn Child



The Neonate

- Neonate- birth to 1 month of age



APGAR test at 1 minute and again 5 minutes after birth

Subtest	0	1	2
A ppearance (Body color)	Blue and pale	Body pink, but extremities blue	Entire body pink
P ulse (Heart rate)	Absent	Slow—less than 100 beats per minute	Fast—100 to 140 beats per minute
G rimace (Reflex irritability)	No response	Grimace	Coughing, sneezing, and crying
A ctivity (Muscle tone)	Limp and flaccid	Weak, inactive, but some flexion of extremities	Strong, active motion
R espiration (Breathing)	No breathing for more than 1 minute	Irregular and slow	Good breathing with normal crying



Brazelton Neonatal Behavioral Assessment Scale (NBAS)

- Rates neonates on 27 items
- Receives rating of worrisome, normal, or superior
- Most effective if given at one day and a week later
- <https://www.youtube.com/watch?v=tqc8gKuXs3s>

Low birthweight

- Low birth weight (LBW) – less than 5.5 pounds (2,500 grams)
 - Pre-term
 - Small for date
- Very low birth weight – less than 3.3 pounds (1,500 grams)
- Extremely low birth weight – less than 2.2 pounds (1,000 grams)

The Preterm Neonate

- Born between 23-37 weeks
- Why come out early?
 - The environment is not suitable for fetus any longer
 - High in stress; teratogens; age of mother; cervical/uterus abnormality
 - In developing countries- **malnourishment, poor prenatal care**
 - In developed countries- **maternal cigarette smoking, use of alcohol, and other drugs during pregnancy, maternal age (>40 and <17)**



Low Birth Weight is a Risk Factor

- ❑ Low-birth weight neonates are at high risk for death
- ❑ Why?
- ❑ Weak immune systems
- ❑ Nervous system under-developed-sucking difficult
- ❑ Difficulty insulating themselves due to lack of fat
- ❑ Lungs not fully formed
- ❑ High risk for being maltreated or neglected and to receive inadequate care

Low Birth Weight continues to be a Risk Factor in childhood, adolescence and adulthood

- ☐ Why?
- ☐ Premature babies need special attention and stimulation which are often unavailable to them
- ☐ Combination of stress of being born early and correlation with SES and home environment
- ☐ Childhood- Language delays and poor school performance
- ☐ Adolescence- Relatively low intelligence scores and chances of repeating grades
- ☐ Adulthood- Continued low educational attainments, chances of having preterm children.
- ☐ However, a rich and stimulating environment can compensate for some of these ill-effects

Video : <https://www.youtube.com/watch?v=EEPHLC6dMGA>

Test your understanding

- Name a few factors/reasons that supported Miles, enabling him to thrive.

Zeskind and Ramey, 1981

- These researchers compared IQ at 36 months- biologically typical children and fetally malnourished children
- In the first few months of life, half of the children in each group had been sent to Daycare (Experimental condition) and the other half had not (Control condition)
- Medical and nutrition was provided to both; but daycare for only one.
- Results: The children who received stimulation (daycare) thrived, even the fetally malnourished ones

table 3.11 Mean Stanford-Binet Scores at 36 Months
for Children in the Zeskind and Ramey Study

A mean of 100 reflects “average” intelligence.



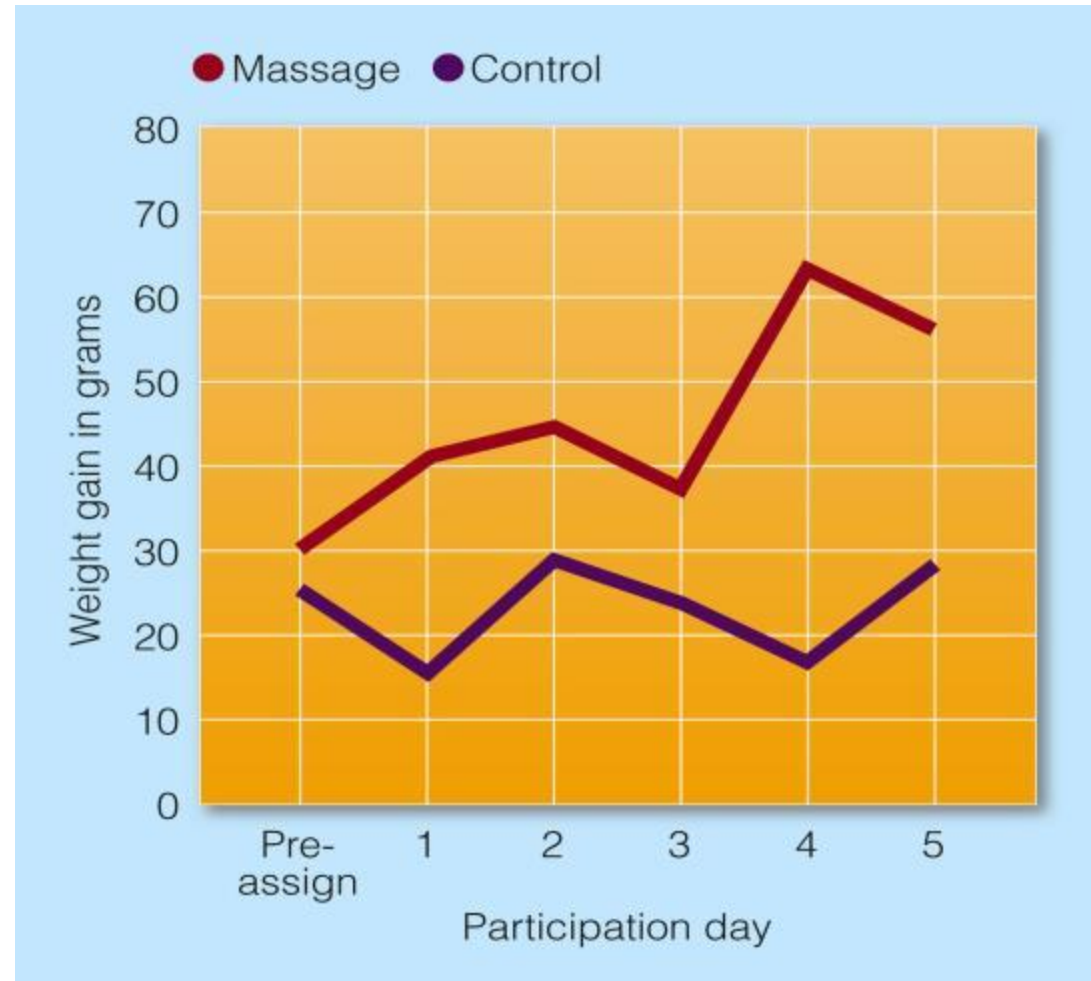
Power of Touch

- Premature infants show substantial gains given consistent physical attention
 - ▣ Kangaroo Care
 - ▣ Therapeutic touch
- Physical contact by mothers releases neurotransmitters that foster brain development and mother-child relationship.



Field et al. (2003)

- 16 preterm babies
- Massage for 15 minutes per day for five days
- Control received no massage
- By DAY FIVE, 53% increase in weight gain!
- Regulate bodily functions-body temperature, breathing, heart rate and sleep wake cycles.
- Mother-neonate bonding



Video:

https://www.ted.com/talks/jane_chen_a_warm_embrace_that_saves_lives?language=en

The Neonate: Sleeping Pattern



- Average- 16-17 hours a day(10 to 21 hours range)
- Sleep wake patterns governed by hunger
- For neonates REM sleep stimulates brain activity

The Neonatal Reflexes

- Infants lack control of their bodies, but these reflexes make responses possible to environmental input.

Reflex	Stimulation	Response	Disappears by
Stepping	Hold baby under arms with feet touching floor	Makes stepping motions	2 months
Moro	Dip downward suddenly, or loud sound	Arch back, extend arms and legs outward, bring arms together swiftly	3 months
Babkin	Press and stroke both palms	Mouth opens, eyes close, head tilts forward	3 months
Sucking	Object or substance in mouth	Sucking	4 months
Rooting	Touch on cheek or mouth	Turn toward touch	4 months
Grasping	Object placed in palm	Hold tightly	4 months
Swimming	Baby is immersed in water	Holds breath, swims with arms and legs	4 months
Babinski	Stroke sole of foot	Foot twists in, toes fan out	8 months

- Can you guess the function of each reflex?
- Why do these disappear? Is that a good thing?

<https://www.youtube.com/watch?v=0V4x0iQODTk>

Neonatal Senses

- Which sense develops first?
- **Touch** is the earliest sense to develop
- By 7 months gestation, all body parts respond to touch
- Neonates feel pain much like adults, contrary to what physicians mistakenly believed previously.
- **Smell and Taste**
- Taste is well developed in the womb
- Can you recall some examples from the **Fetal Learning Ted talk**??
- By 4 days after birth, newborns discriminate the scent of their mother's breast milk from others'

Neonatal Senses

- **Hearing**

- Neonates prefer mother's language to foreign languages

(Vouloumanos et al., 2010)

- Prefer high pitched voices to low pitched voices
- Recognize what they heard while in the womb-
- Dr. Seuss's "Cat in the Hat" OR even the theme of soap operas mothers watched

- **Sight**

- Least developed at birth
- Visual acuity best at a distance of 8-14 inches
- Faces are preferred over any other stimuli
- Neonatal Imitation:
- https://www.youtube.com/watch?v=qSRFvE0Z8Wg&ab_channel=CarbonByGravity



Caring for the Neonate

- Breast Feeding
- Breast feeding practices across cultures
- Watch REVEL video “ Breast Feeding Practices across cultures”
- Similar reasons for breastfeeding across cultures
- What are some benefits mentioned in the video?

Benefits of breast feeding

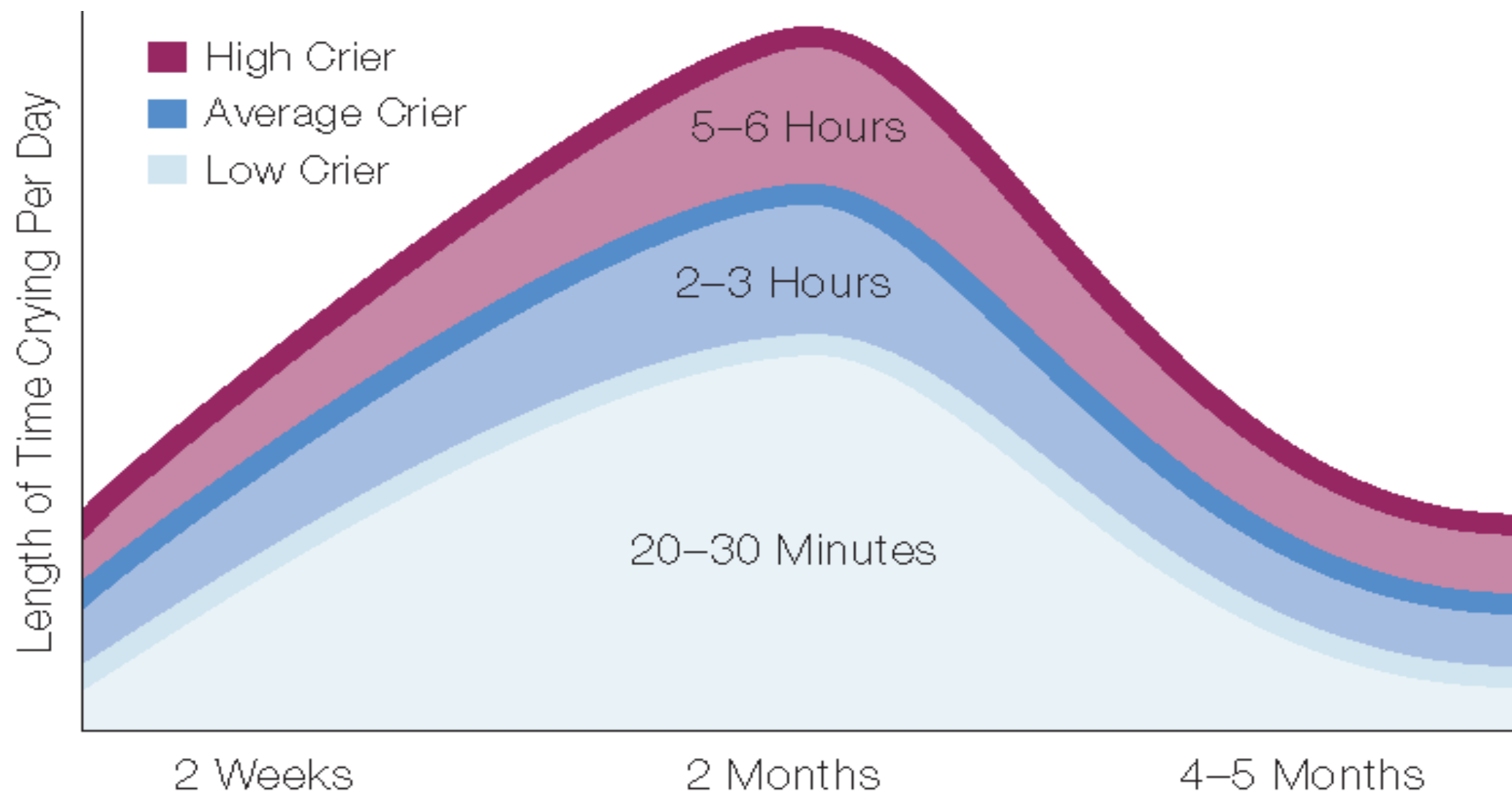
- Disease protection
- Cognitive development
- Nutrients in breast milk promote early brain development
- Positive effects continue till adulthood- greater occupational success and earnings
- Reduced obesity
- Important for developed countries with high rates of obesity
- Better health in childhood and adulthood
- Release of oxytocin- helps bonding, and mother's health
- Return to pre-pregnancy weight
- Strengthens mother's bones, reduces risks for ovarian and breast cancer
- How long to breastfeed?
- WHO recommends 2 years- solid foods introduced at 6 months.



Cognitive benefits of Breast feeding: Correlation or Causation?

- Mothers who choose to breast feed tend to be better educated than mothers who do not.
- Is breast milk really the reason for children's superior cognitive performance?
- Watch Revel Video
"Breast feeding benefits: Separating correlation and causation"

Crying



Soothing

- Swaddling babies has been shown to reduce crying
- Soothing a crying neonate includes
 - Lifting baby by shoulder, soothing repetitive movements, soothing sounds, and distraction
- Common theme is new source of stimulation



Postpartum depression

- Series of depressive episodes appearing within four weeks.
- About 10% of new mothers across 11 countries (Oates et al., 2004)
- Fathers experience it as well
- Different from normal hormonal changes post birth- may last till 6 months after birth
- Genetic vulnerabilities
- Lack of social support
- Infant's developmental problems- irritable, attachment issues
- What type of effects are these?- passive, active, evocative?
- But even so, depressed mothers display lesser talking, touching and smiling with infants (Righetti-Veltema et al., 2002)
- Psychotherapy, Exercise both before and after pregnancy

Are we doing enough to support new mothers and families make the transition?

- WATCH this Video: Maternity leave
- https://www.ted.com/talks/jessica_shortall_how_america_fails_new_parents_and_their_babies